MANJINDER SINGH ROOPRAI

An engineer who builds platforms + automates infrastructure + writes production-grade systems
Senior Technologist | Backend & Platform Engineering | Cloud, DevOps & Generative AI
LinkedIn | **Portfolio** | **GitHub**

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years building scalable backend systems and cloud-native platforms on AWS. Extensive experience delivering enterprise-grade microservices, Kubernetes (EKS) infrastructure, and CI/CD automation for Apple production systems.

Proven track record of engineering end-to-end deployment platforms, including one-click Kubernetes provisioning, improving deployment efficiency, reliability, and system scalability. Strong expertise in platform engineering, DevOps practices, observability (Prometheus, Grafana, Splunk), and distributed system design.

Hands-on experience with Generative AI integrations, Spring AI, and RAG-based architectures, combined with deep knowledge of the full software development lifecycle. Recognized for driving automation, reducing operational overhead, and improving developer productivity in high-scale environments.

PROFESSIONAL EXPERIENCE

Senior Technologist (Backend, Cloud & Platform Engineering) *Jun 2025 – Present*

Sherlog-Assistant MVP | Apple
Infosys Ltd | Bangalore, India

AI-powered Kubernetes observability solution leveraging Apple's Floodgate LLM technology for intelligent anomaly detection and automated root cause analysis across enterprise-scale deployments.

- **Architected and developed** end-to-end AI-powered observability platform using Java 21, Spring AI, and Apple Floodgate LLM, achieving 95% anomaly detection accuracy with 2-3 second analysis latency
- **Engineered intelligent monitoring system** with real-time Kubernetes log streaming, event aggregation, and context-aware anomaly detection, reducing MTTR by 60-80% through automated root cause analysis
- **Built scalable microservices architecture** supporting 100+ pods with minimal resource usage, implementing heuristic fallback mechanisms for 99.9% system reliability
- **Designed comprehensive REST API and web dashboard** with mobile-responsive UI, enabling seamless integration with existing monitoring infrastructure and multi-channel alerting systems
- **Implemented enterprise-grade security** with minimal RBAC permissions (read-only), audit logging, and data encryption support, ensuring compliance with Apple's security standards
- **Delivered measurable business impact** with projected 650%+ ROI in first year, $850K annual cost savings through reduced incident response time and prevented outages
- **Led proof-of-concept development** from inception to completion in 4 weeks, creating comprehensive documentation, performance benchmarking, and integration with sample applications
- **Established deployment strategy** using Helm charts, Docker containers, and Kubernetes-native design for seamless production rollout across multiple environments

Senior Technologist (Backend, Cloud & Platform Engineering) *Dec 2025 – Present*

Paragon (Support.apple.com) | Apple
Infosys Ltd | Bangalore, India

Apple's next-generation support content delivery platform — scalable Spring Boot microservices managing localised content & real-time pricing across all Apple devices.

- **Engineered end-to-end cloud deployment platform on AWS EKS using Infrastructure as Code (CloudFormation) and CI/CD automation**, enabling fully automated, repeatable **one-click Kubernetes provisioning** across environments
- Designed and implemented **Helm-based Kubernetes deployment architecture** for Dev/IT/UAT/STAGE/Prod with environment-driven configuration and scalability

- Built **high-availability microservices infrastructure** using Redis caching, CDN strategies, and circuit breaker patterns to meet strict SLA requirements
- Improved deployment efficiency and reduced manual intervention by **~70% through CI/CD and infrastructure automation**
- Applied **SRE principles (availability, latency, fault tolerance)** to ensure production-grade reliability and performance

Specialist Programmer (Backend, Cloud & Platform Engineering)

Jun 2019 – Dec 2025

Peloton (Apple Online Store) | Apple (Onsite: Burnaby, CA Mar 2023 – Sep 2024)

Infosys Ltd | Bangalore, India

Enterprise Spring Boot + Angular platform monitoring and managing AWS instances across Apple's global data centre footprint.

- **Onsite at Apple Burnaby HQ:** Collaborated directly with Apple engineering teams, gathering infrastructure requirements and delivering cloud deployment & monitoring strategy.
- Led design and implementation of EKS-based cloud infrastructure and CI/CD pipelines for 4 enterprise applications across **high-traffic, distributed AWS environments**
- Built a **one-click deployment system** covering infrastructure provisioning, build, and deployment, significantly improving release velocity
- Owned **end-to-end release lifecycle (Dev/Test/Prod)**, ensuring **zero-downtime deployments and rollback strategies**
- Developed internal platform tools including AWS Inventory & Cost Optimization system, EKS Binary Comparison tool, Config Comparison, Aws Cost Automation, process Inventory, Aws Summary etc and Splunk dashboards for observability and monitoring
- Engineered **centralized Peloton AWS SDK client (PASC)** and service communication layer (PACS) for scalable EKS interactions
- **Led Multiple Applications Design and Development:** Peloton, Peloton AOS Service Client, Peloton AWS Service Client, Peloton AWS Cost Explorer, Peloton Report Generator and Peloton Monitoring
- Reduced **MTTR through proactive monitoring, alerting, and root-cause analysis workflows**
- Collaborated directly with Apple engineering teams onsite to define **cloud architecture and infrastructure strategy**
- Mentored engineers on **Kubernetes, AWS, CI/CD, and Backend engineering best practices**

Senior Software Engineer (Backend Systems)

Dec 2016 – Jun 2019

Templar | Tower Hill Insurance

Mindtree Ltd | Bangalore, India

The Insurance Billing System (IBS) provides the ability to process and track all policy receivables and payables. Unlimited options for the timing and method of billing and collection of receivables were available through client-defined tables. All current and historical billing and payment information is available online.

- Developed and maintained key modules such as **Refund, Disbursement, Policy Locks, and Doc Director.**
- **Monitored UAT (User Acceptance Testing) and managed build processes to ensure smooth deployments.**
- I**mplemented and optimized batch jobs** for various modules, ensuring performance and reliability.
- Maintained the system by **implementing necessary changes, documenting updates, and ensuring compliance with business requirements.**
- Collaborated with **stakeholders to understand workflow and assess the business impact of system changes.**

Software Engineer

Jun 2016 – Dec 2016

Esurplus | AIG

Mindtree Ltd | India

- Built automation solutions for data processing, reporting, and email workflows using Java
- Reduced manual effort and improved operational efficiency through automated reporting pipelines

GENERATIVE AI & INNOVATION

- 2+ years hands-on experience with Generative AI and Spring AI framework, building AI-assisted features integrated into Apple enterprise applications.
- Proficient with Apple-internal AI tooling:Apple Claude code, Apple CodeGenius, Apple Interlinked, Apple Roo Code.

- Experience with IBM Watson Orchestrate, Claude, ChatGPT, Google Gemini, Google AI Studio, and DeepSeek for enterprise AI workflow automation.
- Implemented MCP (Model Context Protocol) server integrations and vector database (ChromaDB, MariaDB) solutions for RAG-based AI applications.

CORE COMPETENCIES

- **Languages:** Java, Python, Groovy, SQL, Bash, Ansible
- **Backend & Systems:** Spring Boot, Spring Batch, Spring Data JPA, Apple Aluminium framework, Hibernate, Apache Camel, Kafka, ActiveMQ, Hazelcast, Redis
- **Cloud & DevOps:** AWS (EKS, EC2, S3, IAM, CDN, EFS, CloudFormation), Kubernetes, Helm, QuikKube, Kind, Docker, CI/CD (Jenkins), Apple Rio, Terraform (working knowledge), GitOps (ArgoCD, Flux – exposure), Infrastructure as Code (IaC)
- **AI / ML:** Spring AI, RAG, MCP, Agentic AI, Claude Code, Apple CodeGenius, Apple Interlinked, Apple Roo Code, IBM Watsonx, ChatGPT, Gemini, DeepSeek, Ollama, Docker model hub
- **Frontend:** Angular 17, Bootstrap, Apple Bricks
- **Observability:** Splunk, Prometheus, Grafana, OpenTelemetry, Spring Actuator
- **Databases:** Oracle SQL, PostgreSQL, MySQL, ChromaDB / MariaDB (VectorDB)
- **Security:** HashiCorp Vault, AWS IAM, Spring Security, OAuth2, Apple Narrative, Apple Whisper, Apple Idms, Apple connect

CERTIFICATIONS

- **Claude Certified Architect - Professional** — *Oct 2026*
- **Claude Certified Developer - Foundations** — *Oct 2026*
- **Claude Certified Architect - Foundations** — *Sep 2026*
- **Claude Certified Associate - Foundations** — *Sep 2026*
- **AWS AI Practitioner** — Infosys / AWS — *Jun 2025*
- **Spring Boot Developer** — Infosys — *Jun 2025*
- **Angular 17** — Infosys — *Mar 2024*
- **Generative AI Capsule Course** — Infosys — *Feb 2024*
- **Full Stack AWS Developer Associate** — Infosys — *Nov 2023*
- **AWS Cloud Practitioner** — Amazon Web Services — *Sep 2019*
- **Spring Boot Professional** — Industry — *May 2018*
- **Microsoft Azure** — Microsoft — *Mar 2017*

AWARDS & RECOGNITION

- Infosys **Insta award** — *Jun 2026*
- Infosys **Rise award - Business Ninja** — *Oct 2025*
- Infosys **Insta award** — *Sep 2024*
- Infosys **Insta award** — *Oct 2024*
- Infosys **Award of Excellence** — *Jun 2023*
- Infosys **Zenith Award – Most value player** — *Jul 2020*
- Mindtree **Outstanding Performer Award** — *Nov 2019*
- Mindtree **Mastermind** Award — *Sep 2018*
- Mindtree **Unstoppable** Award — *Jun 2017*

EDUCATION

Bachelor of Technology — Information Technology | *Mar 2012 – Jun 2016*
Chandigarh Engineering College | Mohali, Punjab

INTERESTS

Digital Creator
Verses